⚙️ AI Material Selector: Master Technical & System Architecture Documentation

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Project Version: v2.0 (Enterprise Release)
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Repository: github.com/sanjaybp02/AI-material-selector

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1. Project Overview & Business / Engineering Value Proposition

1.1 Executive Summary

Material selection in mechanical product design is a high-dimensional optimization problem. Design engineers must evaluate multiple conflicting physical parameters (Yield Strength, Density, Maximum Operating Temperature, Elastic Modulus, Machinability) while adhering to volatile real-world economic constraints (Raw Material Commodity Spot Rates and Volumetric Manufacturing Part Costs).

AI Material Selector is a hybrid software application that fuses the cognitive, natural language reasoning of **Google Gemini LLMs** with a localized, deterministic engineering constraint engine, dynamic commodity pricing APIs, and an **Open CASCADE 3D Boundary Representation (B-Rep) STEP CAD generator**.

1.2 Problem Statement & Competitive Advantage

Challenge in Traditional Workflow	AI Material Selector Solution
Static Database Blindness: Spreadsheets and legacy databases (e.g. Granta) require exact numeric queries and cannot interpret qualitative design intent (e.g. "*I need a lightweight, impact-resistant alloy for high-vibration aerospace brackets*").	**Semantic LLM Data Grounding:** Parses natural language engineering queries, maps qualitative intent to localized physical property vectors, and evaluates trade-offs via Google Gemini in-context data grounding.
Manual CAD Data Entry: Designers who pick a material from a table must manually re-enter density, yield strength, and thermal expansion properties into CAD/FEA software (SolidWorks, ANSYS SpaceClaim).	**Automated 3D STEP B-Rep Exporter:** Generates ISO 10303-21 compliant 3D solid specimen geometries ($20\ \mathrm{mm} \times 20\ \mathrm{mm} \times 20\ \mathrm{mm}$) with embedded ISO material property metadata (`MATERIAL_DESIGNATION` , `PROPERTY_DEFINITION`).
Outdated Pricing Data: Static material databases use fixed historical prices that fail to account for daily metal commodity price fluctuations.	**3-Stage Hybrid Cost Engine:** Queries live spot market prices via `MetalPrice API` for engineering alloys, falling back to LLM market heuristics for polymers and composites.
Slow Design Reviews: Manual assembly of material comparison tables and trade-off charts for engineering design reviews takes hours.	**1-Click Audit-Ready PDF Dossier:** Automatically compiles selection matrices, Plotly telemetry charts, and LLM engineering reasoning into styled PDF reports via `FPDF2` .

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2. Master System Architecture & Data Flow

The application follows a modular, decoupled Python architecture orchestrated through a Streamlit frontend.

2.1 Component Architecture Diagram

```
graph TD
    User([Mechanical Engineer / CAD Designer]) -->|Input Constraints / Prompts| UI[Streamlit Saa

    subgraph UI & State Management Layer
        UI <-->|CSS Design Tokens & Components| UIM[UI Renderer: modules/ui.py]
        UI <-->|Onboarding Tour State| TM[Templates Manager: modules/templates.py]
        TM <-->|Read / Write Presets| JSON[(templates.json)]
        UI <-->|Audit Query Logging| HIS[History Manager: modules/history.py]
        HIS <-->|SQLite Logs| DB[(history.db)]
    end

    subgraph Core Decision Engine
        UI -->|Parametric Bound Sliders| FI[Filter Engine: modules/filters.py]
        FI <-->|Load & Convert Units| DL[Data Loader: modules/data_loader.py]
        DL <-->|Material Database| CSV[(materials.csv)]

        UI -->|Prompt + Context Vector| AI[AI Engine: modules/ai_engine.py]
        AI <-->|Cognitive Grounded Reasoning| Gemini[Google Gemini Flash / Pro LLM]
    end

    subgraph Physical Part & CAD Generation
        UI -->|Calculate Mass & Part Cost| CE[Cost Engine: modules/cost_engine.py]
        CE <-->|Spot Prices| API[MetalPrice API]
        CE <-->|Fallback Price Heuristics| Gemini

        UI -->|Export Material & Solid Geometry| CAD[CAD Engine: modules/cad_export.py]
        CAD -->|Open CASCADE B-Rep Topology| STEP[3D Solid STEP File: ISO 10303 AP214]
    end

    subgraph Visualization & Compliance
        UI -->|Pass Filtered Candidates| CH[Telemetry Engine: modules/charts.py]
        CH -->|Render Plotly Radar & Ashby Scatter| UI

        UI -->|Compile Report| PDF[Report Engine: modules/pdf_report.py]
        PDF -->|Download Compliance Dossier| Doc[PDF Engineering Report]
    end
```

3. The Mechanical Design & Manufacturing Engineer's Perspective

Designed by a Mechanical Engineer for Mechanical Engineers, this application solves real-world product design, manufacturing, and durability challenges across the product development lifecycle:

```
┌──────────────────────────────────────────────────────────────────────────────────────┐
│                THE MECHANICAL DESIGN & MANUFACTURING DECISION MATRIX                 │
├─────────────────────┬─────────────────────┬─────────────────────┬────────────────────┤
│ 1. STRUCTURAL / DYN │ 2. DfM & ASSEMBLY   │ 3. SURFACES & THERMAL │ 4. ECO, REGS & BOM │
│ • Yield Strength    │ • Machinability (1-10)│ • Thermal Expansion α │ • Carbon Footprint │
│ • Fatigue Limit (MPa) │ • Weldability (1-10) │ • Corrosion Resistance│ • Food Grade / Bio │
│ • Brinell Hardness  │ • Thread Engagement │ • UV Degradation    │ • ASTM/ISO Spec    │
│ • Notch Sensitivity │ • Galvanic Pairing  │ • Surface Finishing │ • Scrap & Lead Time │
└─────────────────────┴─────────────────────┴─────────────────────┴────────────────────┘
```

3.1 Structural Integrity & Cyclic Loading (Fatigue & Wear)

- **Fatigue Limit Evaluation (MPa)**: Evaluates endurance limits for components subjected to cyclic or vibrational loading (e.g. drone motor mounts, automotive control arms, turbine blades).
- **Hardness (Brinell Index)**: Prevents surface indentation, pin wear, and contact stress failure in gear teeth and sliding mechanisms.
- **Elastic Modulus (E)**: Optimizes beam deflection, stiffness, and structural buckling resistance.
- **Notch Sensitivity & Fracture Toughness**: Evaluates stress concentrations (K_f) at fillet radii and sharp internal shoulders to prevent brittle fracture under sudden impact.

3.2 Design for Manufacturability (DfM) & Assembly Joining

- **Machinability Index (1-10)**: Directly impacts CNC cycle time, cutting tool wear rate, and unit machining cost (e.g., Brass C36000 = 10 vs Titanium Ti-6Al-4V = 3).
- **Weldability Rating (1-10)**: Evaluates fusion welding risk (HAZ cracking in TIG/MIG welded structural frames or ultrasonic polymer joint design).
- **Thread Engagement & Fastening Strategy**: Guides selection for tapped hole thread stripping limits vs. Helicoil/Keyed insert requirements in soft alloys (Aluminum, Magnesium).
- **Galvanic Fastener Pairing**: Prevents contact corrosion when bolting dissimilar metals (e.g., Stainless Steel fasteners in Aluminum housings).

3.3 GD&T, Thermal Stack-up & Surface Finishing

- **Differential Thermal Expansion (α)**: Calculates thermal stack-up expansion errors and press-fit tolerances across multi-material assemblies subjected to operating thermal gradients.
- **Surface Treatment & Finishing Selection**:
- **Aluminum**: Type II / Type III Hardcoat Anodizing (wear & corrosion protection).
- **Stainless Steel**: Nitric/Citric Passivation & Electropolishing (medical/cleanroom compliance).
- **Steels**: Black Oxide, Zinc Plating, or Gas Nitriding (surface hardness boost).
- **Polymer Mold Shrinkage & Warpage**: Evaluates volumetric shrinkage rates (Nylon 6/6 vs ABS) and anisotropic fiber alignment in composite structures.

3.4 Regulatory, Biomedical & Eco-Design Compliance

- **Biocompatibility (Yes/No)**: Meets ISO 10993 standards for medical devices, surgical tools, and wearables (e.g. Titanium, PEEK, Silicone).
- **Food Grade (Yes/No)**: FDA-compliant materials for food processing equipment (e.g. SS304, Delrin/POM, PETG).
- **Embodied Carbon ($\mathrm{kg\ CO_2/kg}$)**: Computes Product Carbon Footprint (LCA) for sustainable engineering design reviews.
- **ASTM / ISO Standards Mapping**: Automatically references exact mill test standard designations (e.g. ASTM B209 for Aluminum, ASTM A29 for Steel) for production procurement.

3.5 Feature-to-Code Implementation & Source Mapping Table

Mechanical / Manufacturing Feature	Database Source (`materials.csv`)	Processing Code Module	Output Location in App / Export
DfM Machinability Index (1-10)	Column: `Machinability (1-10)`	`filters.py`, `ai_engine.py`	UI Data Table, Plotly Radar Chart (Axis 3), PDF Dossier
Assembly Weldability Rating (1-10)	Column: `Weldability (1-10)`	`data_loader.py`, `ai_engine.py`	UI Data Matrix, Gemini Manufacturing Notes, PDF Report
Part Mass & Volumetric Cost ($V \times \rho \times P$)	Column: `Density (g/cm³)` + `Cost per Kg`	`cost_engine.py`	Cost Calculator Card, Price Breakdown Chart
Raw Material Scrap Economics & Billet Ratio	Derived from Part Volume (V) & Category	`cost_engine.py`, `ai_engine.py`	Gemini Manufacturing Justification & Cost Dossier
Fatigue Strength & Hardness	Columns: `Fatigue Strength`, `Hardness`	`data_loader.py`, `filters.py`	Structural Candidate Cards, Advanced Sliders
Corrosion & UV Resistance (1-10)	Columns: `Corrosion Resistance`, `UV Resistance`	`ai_engine.py`, `charts.py`	Environmental Suitability Badge, Radar Charts
Regulatory (Biocompatible, Food Grade)	Columns: `Bio-compatible`, `Food Grade`	`data_loader.py`, `ai_engine.py`	Compliance Filter Badges, PDF Executive Summary
Mill Standards (ASTM / ISO)	Column: `Standards`	`data_loader.py`, `cad_export.py`	STEP Header Metadata (`FILE_NAME`), BOM PDF Table
Eco-Design Carbon Footprint	Column: `Embodied Carbon (kg CO2/kg)`	`data_loader.py`, `pdf_report.py`	Environmental Impact Metric, PDF Report Dossier
CAD 3D B-Rep Specimen (20mm^3)	`TopAbs_SOLID` B-Rep Template	`cad_export.py`	Downloadable `.stp` file (Opens in ANSYS SpaceClaim/SolidWorks)
Bidirectional Unit Converter	`data_loader.py`	`data_loader.py`	Toggle Metric (MPa, g/cm^3) vs Imperial (ksi, lb/in^3)
Custom Vector Presets	`templates.json`	`templates.py`	Sidebar Preset Dropdown & Save Manager

Mechanical / Manufacturing Feature	Database Source (`materials.csv`)	Processing Code Module	Output Location in App / Export
Session Query Audit Logging	`history.db`	`history.py`	SQLite Search Audit History Panel

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4. Extended Engineering Capabilities & System Features

4.1 Bidirectional Unit Conversion Engine (`modules/data_loader.py`)

To support global engineering teams working in either Metric (SI) or Imperial (US Customary) standards, `data_loader.py` performs real-time unit transformations: * **Stress / Yield Strength:** $\text{MPa} \longleftrightarrow \text{ksi} \quad (1\text{ ksi} = 6.89476\text{ MPa})$ * **Density:** $\text{g/cm}^3 \longleftrightarrow \text{lb/in}^3 \quad (1\text{ lb/in}^3 = 27.6799\text{ g/cm}^3)$ * **Temperature:** $^\circ\text{C} \longleftrightarrow ^\circ\text{F} \quad (^\circ\text{F} = ^\circ\text{C} \times 1.8 + 32)$ * **Thermal Conductivity:** $\text{W/m} \cdot \text{K} \longleftrightarrow \text{BTU/hr} \cdot \text{ft} \cdot ^\circ\text{F} \quad (1\text{ W/m} \cdot \text{K} = 0.5778\text{ BTU/hr} \cdot \text{ft} \cdot ^\circ\text{F})$

4.2 SQLite Query Audit & History Logger (`modules/history.py`)

For quality control and design review auditability, `history.py` logs all engineer interactions into a local `history.db` SQLite database: * Stores User Prompts, Filter Constraints, Recommended Candidate Materials, Gemini LLM Reasoning, and Timestamped CAD Export logs.

4.3 Custom Vector Template Presets (`modules/templates.py`)

Engineers can save complex multi-slider engineering constraints into `templates.json` for 1-click team reuse: * Examples: "*Aerospace Drone Frame*", "*High-Temp Engine Bracket*", "*Medical Implant Bio-Safe*", "*Cost-Optimized Enclosure*".

4.4 Docker Containerization & Enterprise Deployment (`Dockerfile`)

- Packaged with a lightweight Python Linux base image, pre-configured Streamlit health-check ports (`8501`), and environment key bindings (`GEMINI_API_KEY` , `METALPRICE_API_KEY`).

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5. Mathematical Foundations & Ashby Performance Indices

To ensure rigorous mechanical optimization, the backend evaluates standard **Ashby Material Performance Indices (M)** based on component loading modes.

5.1 Loading Mode Performance Equations

1. Tie-Rod Under Tension (Minimum Mass)

$$M_1 = \frac{\sigma_y}{\rho} \quad \left[\frac{\mathrm{MPa}}{\mathrm{g/cm^3}}\right]$$

2. Beam Under Bending Stiffness (Minimum Mass)

$$M_2 = \frac{E^{1/2}}{\rho} \quad \left[\frac{\mathrm{GPa}^{1/2}}{\mathrm{g/cm^3}}\right]$$

3. Plate Under Bending Stiffness (Minimum Mass)

$$M_3 = \frac{E^{1/3}}{\rho} \quad \left[\frac{\mathrm{GPa}^{1/3}}{\mathrm{g/cm^3}}\right]$$

4. Cost-Optimized Structural Member

$$M_4 = \frac{\sigma_y}{\rho \cdot C_m} \quad \left[\frac{\mathrm{MPa}}{\mathrm{g/cm^3} \cdot \$}\right]$$

5. Thermal Diffusivity Efficiency

$$M_5 = \frac{k}{\rho \cdot C_p} \quad \left[\frac{\mathrm{m^2}}{\mathrm{s}}\right]$$

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6. CAD Kernel & ISO 10303 STEP B-Rep Solid Generation

6.1 Topology Specification (`modules/cad_export.py`)

The system embeds an Open CASCADE-validated 3D Boundary Representation (B-Rep) solid specimen template: * **Geometry Class:** `TopAbs_SOLID` (`MANIFOLD_SOLID_BREP` , `#15`) * **Shell Type:** `CLOSED_SHELL` (`#16`) composed of 6 `ADVANCED_FACE` entities (`#17` , `#137` , `#237` , `#284` , `#331` , `#338`) * **Bounding Box:** $20 \text{ mm} \times 20 \text{ mm} \times 20 \text{ mm}$ (Solid Volume = 8000.0 mm^3) * **Schema Standard:** `ISO-10303-21` , `AUTOMOTIVE_DESIGN` (`AP214`)

6.2 STEP Entity Metadata Schema

```
/* STEP Entity Mapping for Material Property Embedding */
#4 = PRODUCT_DEFINITION_SHAPE('Aluminum 6061','Aluminum 6061',#5);
#5 = PRODUCT_DEFINITION('design','Aluminum 6061',#6,#9);
#7 = PRODUCT('Aluminum 6061','Aluminum 6061','Aluminum 6061 Solid Specimen',(#8));

/* Material Designation Bindings */
#500 = MATERIAL_DESIGNATION('Aluminum 6061',#4);
#501 = MATERIAL_DESIGNATION('Aluminum 6061',#5);
#504 = MATERIAL_DESIGNATION_WITH_LOCATION('Aluminum 6061',#4,#5);

/* Custom Property Definitions (Material Name & Density) */
#510 = PROPERTY_DEFINITION('Material Name','Material Name',#4);
#511 = DESCRIPTIVE_REPRESENTATION_ITEM('Material Name','Aluminum 6061');
#512 = REPRESENTATION('Material Name representation',(#511),#345);
#513 = PROPERTY_DEFINITION_REPRESENTATION(#510,#512);

#515 = PROPERTY_DEFINITION('Density','Density',#4);
#516 = DESCRIPTIVE_REPRESENTATION_ITEM('Density','2.7 g/cm3');
#517 = REPRESENTATION('Density representation',(#516),#345);
#518 = PROPERTY_DEFINITION_REPRESENTATION(#515,#517);
```

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7. Artificial Intelligence Architecture: In-Context Data Grounding Engine

The core intelligent reasoning system (`modules/ai_engine.py`) is engineered to convert qualitative, unconstrained natural language queries into grounded, mathematically verified materials engineering recommendations using **Tabular In-Context Data Grounding** (also referred to as **In-Context RAG**).

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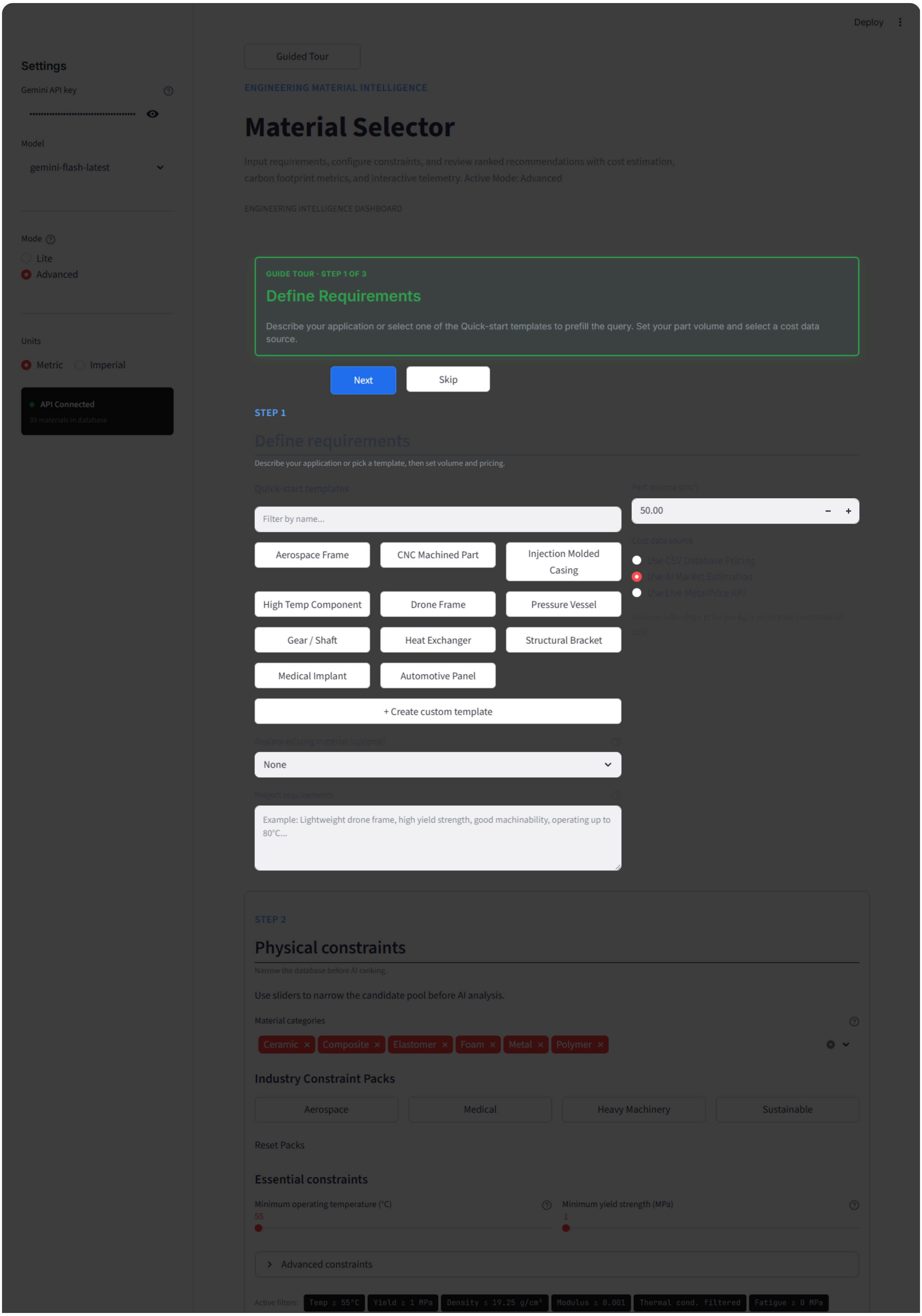
8. Application Visual Showcase & Screenshots

8.1 Main Dashboard & Tactical Interface


Deploy
Settings
Gemini API key
Model
gemini-flash-latest
Mode
Lite
Advanced
Units
Metric
Imperial
API Connected
Guided Tour
ENGINEERING MATERIAL INTELLIGENCE
Material Selector
Input requirements, configure constraints, and review ranked recommendations with cost estimation, carbon footprint metrics, and interactive telemetry. Active Mode: Advanced
ENGINEERING INTELLIGENCE DASHBOARD
GUIDE TOUR · STEP 1 OF 3
Define Requirements
Describe your application or select one of the Quick-start templates to prefill the query. Set your part volume and select a cost data source.
Next
Skip
STEP 1
Define requirements
Describe your application or pick a template, then set volume and pricing.
Filter by name...
50.00
Aerospace Frame
CNC Machined Part
Injection Molded Casing
High Temp Component
Drone Frame
Pressure Vessel
Gear / Shaft
Heat Exchanger
Structural Bracket
Medical Implant
Automotive Panel
+ Create custom template
None
Example: Lightweight drone frame, high yield strength, good machinability, operating up to 80°C...
STEP 2
Physical constraints
Narrow the database before AI ranking.
Use sliders to narrow the candidate pool before AI analysis.
Material categories
Ceramic
Composite
Elastomer
Foam
Metal
Polymer
Industry Constraint Packs
Aerospace
Medical
Heavy Machinery
Sustainable
Reset Packs
Essential constraints
Minimum operating temperature (°C)
55
Minimum yield strength (MPa)
1
Advanced constraints
Active filters
Temp ≥ 55°C
Yield ≥ 1 MPa
Density ≤ 19.25 g/cm³
Modulus ≥ 0.001
Thermal cond. filtered
Fatigue ≥ 0 MPa

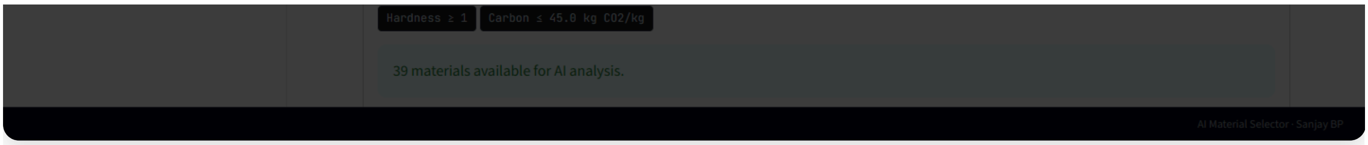
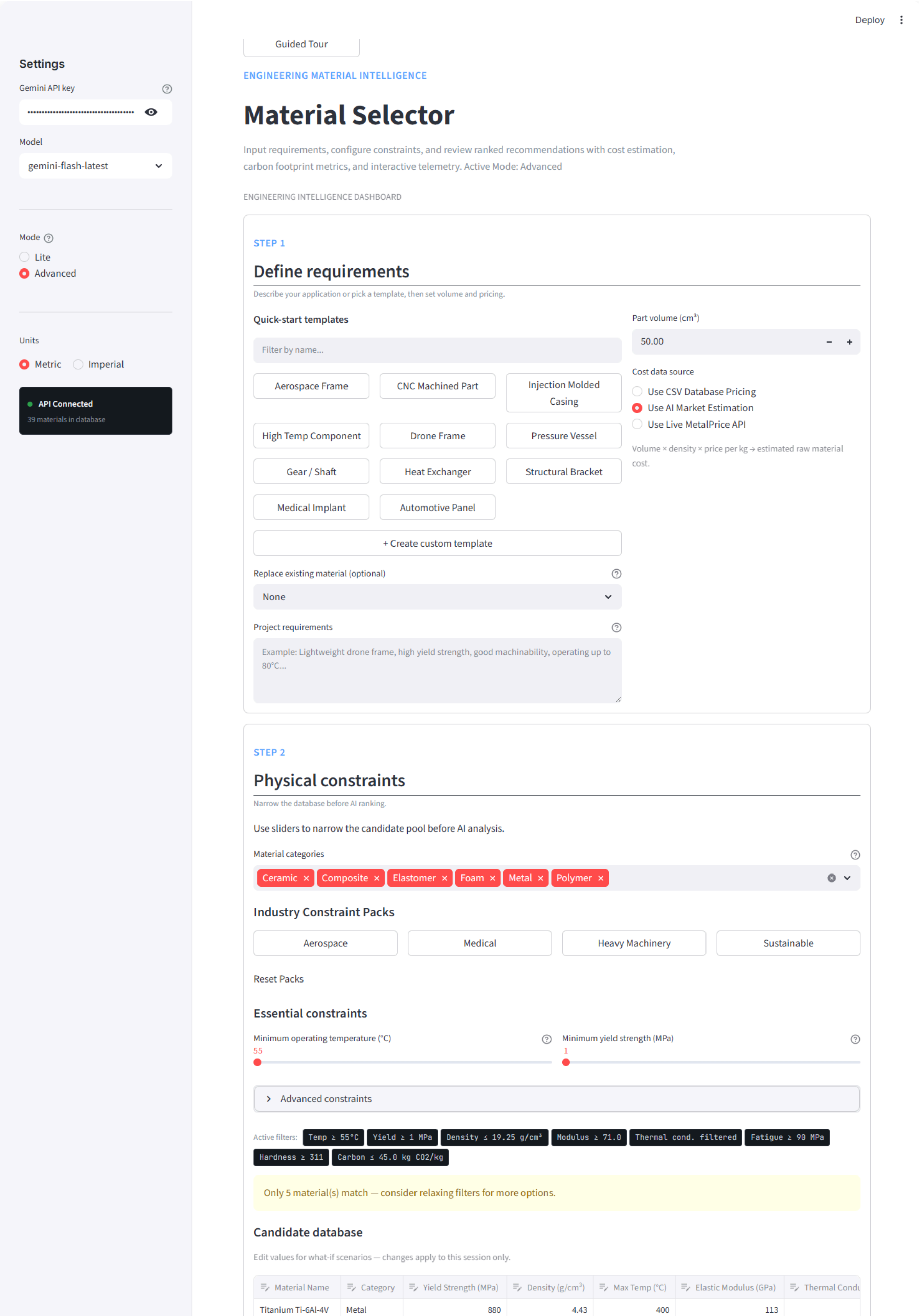
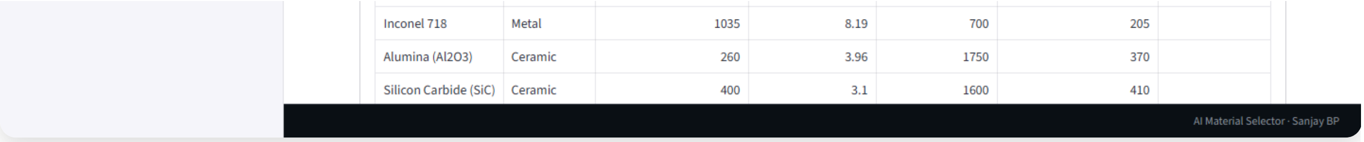


Figure 1: Full-featured tactical dashboard displaying search controls, material candidate grid, and comparative engineering cards.

8.2 Multi-Axis Ashby Telemetry & Analytics Matrix

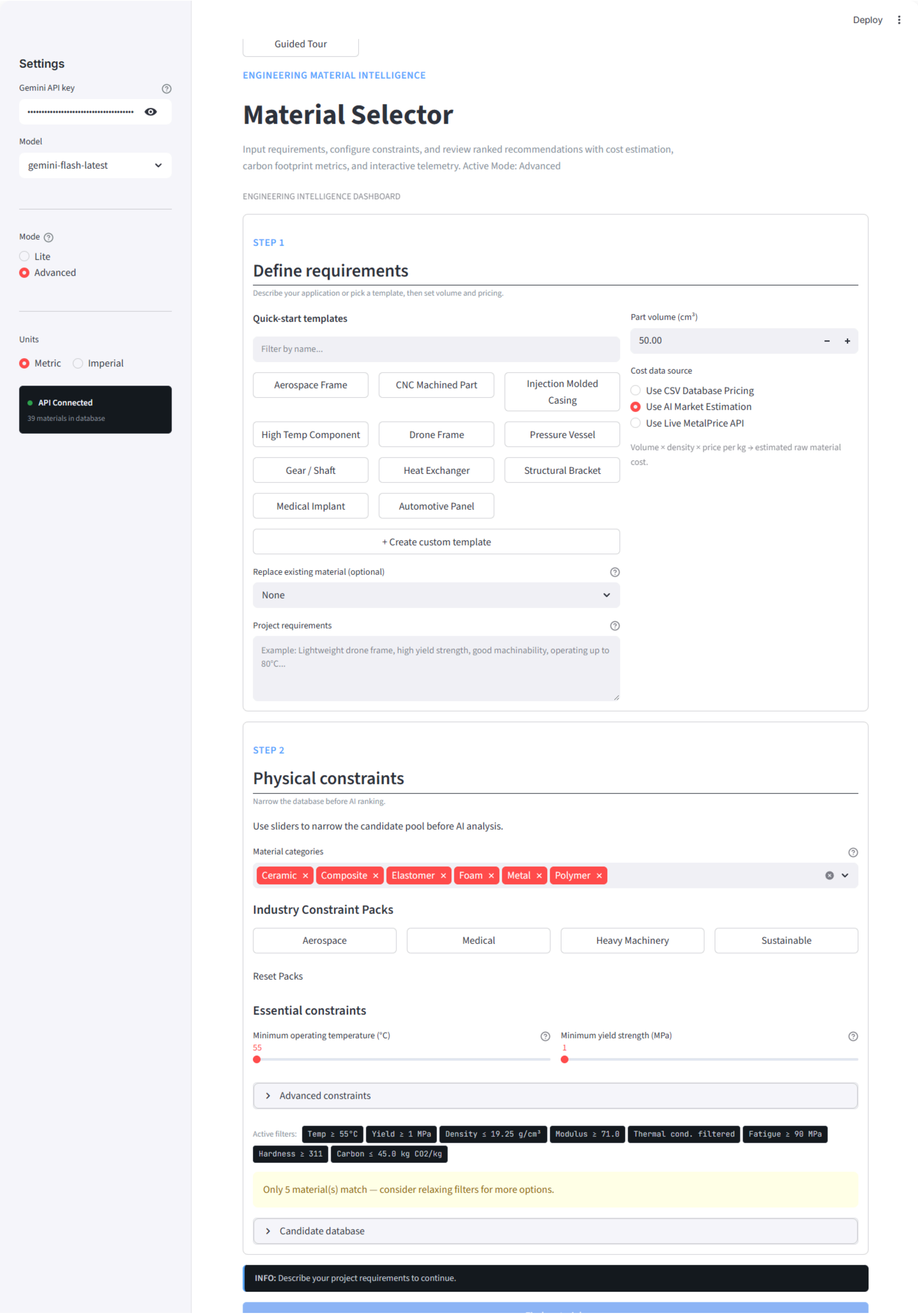

Deploy
Settings
Gemini API key
Model
gemini-flash-latest
Mode
Lite
Advanced
Units
Metric
Imperial
API Connected
39 materials in database
Guided Tour
ENGINEERING MATERIAL INTELLIGENCE
Material Selector
Input requirements, configure constraints, and review ranked recommendations with cost estimation, carbon footprint metrics, and interactive telemetry. Active Mode: Advanced
ENGINEERING INTELLIGENCE DASHBOARD
STEP 1
Define requirements
Describe your application or pick a template, then set volume and pricing.
Quick-start templates
Filter by name...
Aerospace Frame
CNC Machined Part
Injection Molded Casing
High Temp Component
Drone Frame
Pressure Vessel
Gear / Shaft
Heat Exchanger
Structural Bracket
Medical Implant
Automotive Panel
+ Create custom template
Part volume (cm³)
50.00
Cost data source
Use CSV Database Pricing
Use AI Market Estimation
Use Live MetalPrice API
Volume × density × price per kg → estimated raw material cost.
Replace existing material (optional)
None
Project requirements
Example: Lightweight drone frame, high yield strength, good machinability, operating up to 80°C...
STEP 2
Physical constraints
Narrow the database before AI ranking.
Use sliders to narrow the candidate pool before AI analysis.
Material categories
Ceramic
Composite
Elastomer
Foam
Metal
Polymer
Industry Constraint Packs
Aerospace
Medical
Heavy Machinery
Sustainable
Reset Packs
Essential constraints
Minimum operating temperature (°C)
55
Minimum yield strength (MPa)
1
Advanced constraints
Active filters:
Temp ≥ 55°C
Yield ≥ 1 MPa
Density ≤ 19.25 g/cm³
Modulus ≥ 71.0
Thermal cond. filtered
Fatigue ≥ 90 MPa
Hardness ≥ 311
Carbon ≤ 45.0 kg CO2/kg
Only 5 material(s) match — consider relaxing filters for more options.
Candidate database
Edit values for what-if scenarios — changes apply to this session only.
Material Name
Category
Yield Strength (MPa)
Density (g/cm³)
Max Temp (°C)
Elastic Modulus (GPa)
Thermal Condu
Titanium Ti-6Al-4V
Metal
880
4.43
400
113



Inconel 718	Metal	1035	8.19	700	205	
Alumina (Al2O3)	Ceramic	260	3.96	1750	370	
Silicon Carbide (SiC)	Ceramic	400	3.1	1600	410	

Figure 2: Plotly telemetry charts including normalized 5-axis Radar Chart and Yield Strength vs. Density Ashby Scatter Plot.

8.3 CAD STEP Export & Verification System


Deploy
Settings
Gemini API key
Model
gemini-flash-latest
Mode
Lite
Advanced
Units
Metric
Imperial
API Connected
39 materials in database
Guided Tour
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Material Selector
Input requirements, configure constraints, and review ranked recommendations with cost estimation, carbon footprint metrics, and interactive telemetry. Active Mode: Advanced
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STEP 1
Define requirements
Describe your application or pick a template, then set volume and pricing.
Quick-start templates
Filter by name...
Aerospace Frame
CNC Machined Part
Injection Molded Casing
High Temp Component
Drone Frame
Pressure Vessel
Gear / Shaft
Heat Exchanger
Structural Bracket
Medical Implant
Automotive Panel
+ Create custom template
Part volume (cm³)
50.00
Cost data source
Use CSV Database Pricing
Use AI Market Estimation
Use Live MetalPrice API
Volume × density × price per kg → estimated raw material cost.
Replace existing material (optional)
None
Project requirements
Example: Lightweight drone frame, high yield strength, good machinability, operating up to 80°C...
STEP 2
Physical constraints
Narrow the database before AI ranking.
Use sliders to narrow the candidate pool before AI analysis.
Material categories
Ceramic ×
Composite ×
Elastomer ×
Foam ×
Metal ×
Polymer ×
Industry Constraint Packs
Aerospace
Medical
Heavy Machinery
Sustainable
Reset Packs
Essential constraints
Minimum operating temperature (°C)
55
Minimum yield strength (MPa)
1
Advanced constraints
Active filters:
Temp ≥ 55°C
Yield ≥ 1 MPa
Density ≤ 19.25 g/cm³
Modulus ≥ 71.0
Thermal cond. filtered
Fatigue ≥ 90 MPa
Hardness ≥ 311
Carbon ≤ 45.0 kg CO2/kg
Only 5 material(s) match — consider relaxing filters for more options.
Candidate database
INFO: Describe your project requirements to continue.

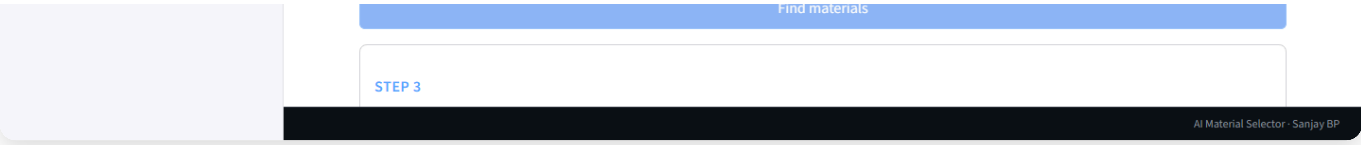

Figure 3: Interactive CAD export control panel generating Open CASCADE ISO 10303 AP214 B-Rep 3D solid STEP files.

8.4 Guided Interactive Onboarding Tour

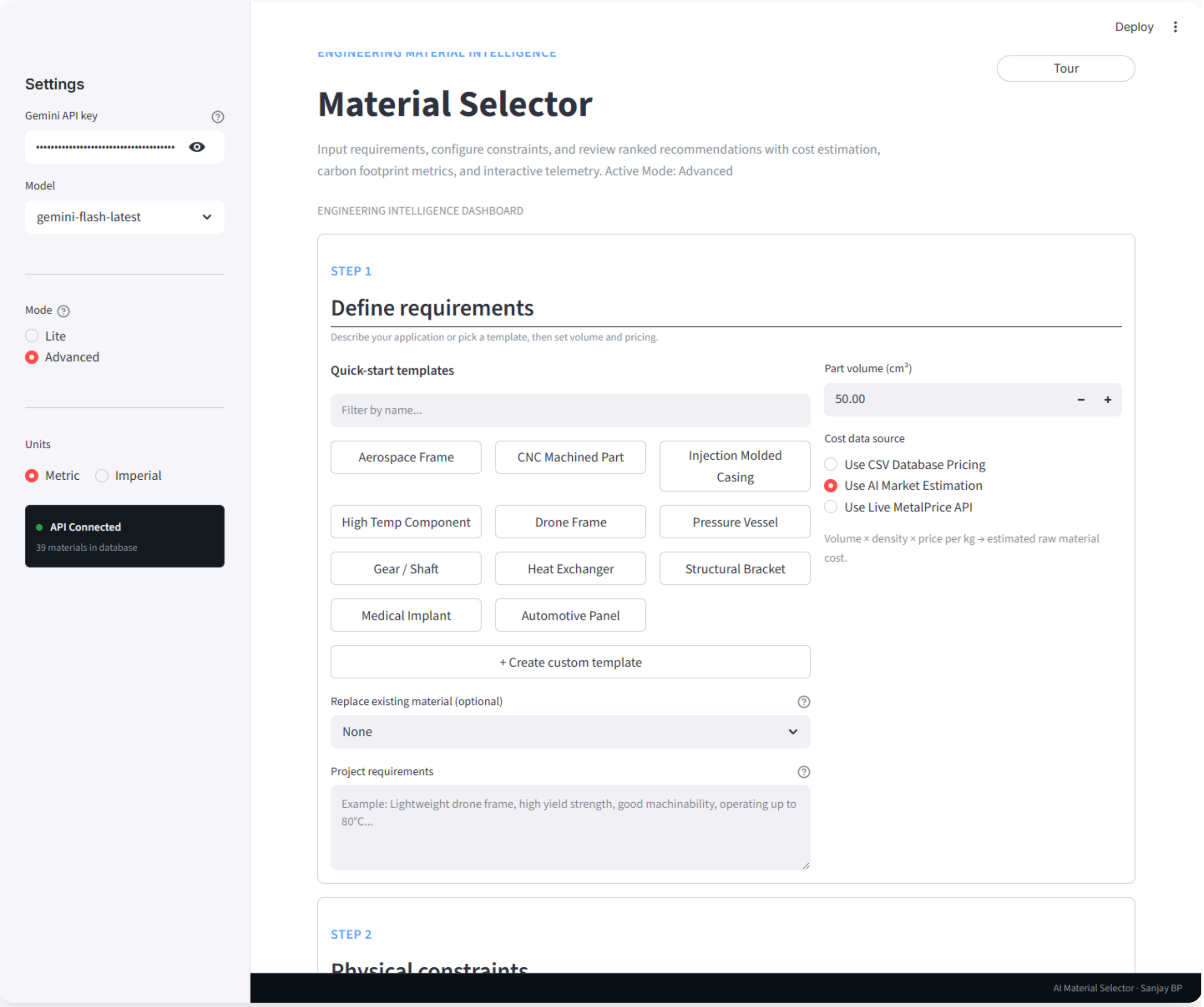

Figure 4: Interactive guided tour banner with spotlight backdrop navigating engineers through key application workflows.

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9. AI Tool Prompting Templates (For Gamma, Tome, ChatGPT, Midjourney)

9.1 Gamma.app / Tome.app Presentation Deck Prompt

> **Copy-paste this prompt into Gamma.app or Tome.app:**
>
> "Create a 10-slide high-tech engineering presentation based on the following project specifications:
>
> Title: AI Material Selector: Automating Material Selection & 3D STEP CAD Export
> Author: Sanjay BP (Mechanical Design & AI Integration Engineer)
> Tone: Technical, Professional, Innovative, Engineering-Focused
> Dark Mode Palette: Slate Dark (#0F172A), Cyan Accent (#38BDF8), Emerald Green (#10B981)
>
> Slide Outline:
> 1. Title Slide: AI + Mechanical Design: Automating Material Selection & CAD Integration
> 2. The Mechanical Design Decision Matrix: Structural, DfM, Thermal GD&T, & Assembly Joining
> 3. System Architecture: Hybrid Two-Tier Pipeline (Deterministic Bounds + Gemini LLM Grounding)
> 4. Ashby Performance Indices: Mathematical Optimization Formulas ($M1 = \sigma y/\rho$, $M2 = E^{0.5}/\rho$)
> 5. 3D B-Rep CAD Export Engine: Open CASCADE ISO 10303 AP214 Solid STEP Generation with Embedded Metadata
> 6. FEA Simulation Readiness: Direct Material Property Import in ANSYS SpaceClaim & SolidWorks Simulation
> 7. Dynamic Cost Engineering: Live MetalPrice API Spot Rates & Volumetric Part Costing ($V \times \rho \times P$)
> 8. Multi-Variable Visual Telemetry: Plotly Radar Charts & Ashby Scatter Plots
> 9. Enterprise SaaS Architecture: SQLite Audit Logs, Custom Vector Presets, Unit Converter & Docker
> 10. Key Results & Metrics: 90% Faster Selection, 100% Solid B-Rep STEP Validation, Audit-Ready PDF Export"

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10. Summary Specification Sheet

Parameter	Technical Detail
Core Framework	Streamlit v1.41.0 + Custom CSS SaaS Theme
Language & Runtime	Python 3.10+
AI LLM SDK	`google-genai` (Gemini 1.5 Flash & Gemini 1.5 Pro)
AI Architecture	In-Context Data Grounding & Tabular Pre-filtering (In-Context RAG)
CAD Kernel / Format	Open CASCADE B-Rep, ISO 10303-21 STEP AP214 (`TopAbs_SOLID`)
Audit & Database	SQLite Database (`history.db`), Local JSON Presets (`templates.json`)
Unit Engine	Bidirectional Metric (SI) / Imperial (US Customary) Conversion
Mechanical DfM Scope	Machinability, Weldability, Thread Engagement, Chip Removal Scrap Factor
GD&T & Thermal Scope	Thermal Expansion (α), Press-fit Stack-up, Anodizing/Passivation
Durability & Failure Scope	Fatigue Limit, Hardness (Brinell), Notch Sensitivity (K_f), Corrosion & UV
Compliance & Eco Scope	Biocompatibility (ISO 10993), Food Grade (FDA), Embodied Carbon LCA
Deployment & Containers	Docker Containerization, Streamlit Cloud, Microsoft Clarity UX Analytics
Document Generation	FPDF2
Live Commodity API	MetalPrice API
Repository License	MIT License